CLAUDE SKILLS ▪ GUIDE

Using pdf-to-markdown with academic-pdf-reading

Accurate equations at lower token cost | 12 October 2026

At a glance

In one sentence: convert equation-heavy PDFs to Markdown once with **pdf-to-markdown**, and **academic-pdf-reading** will read the transcription as exact source, so Claude gets image-verified equations without paying for page images in every session.

1 The trade-off

academic-pdf-reading is accurate because it never takes mathematics from a PDF's text layer. Every sign, index, bar, prime and typeface is read from the page image: a full page view, then an upscaled crop of each line that matters. That accuracy has a cost, paid on every question about an equation and paid again in every new session, because nothing Claude saw in one conversation carries over to the next.

2 Transcribe once, read as source

pdf-to-markdown spends the image-reading effort a single time. Claude transcribes every formula from page images into LaTeX. Scripts then check the syntax, equation tags and page markers, and each displayed equation is proofread side by side against its crop from the PDF. The result is a text-only `NAME.md` plus `NAME_figures.pdf`.

academic-pdf-reading recognises a transcription by its first-line legend and treats it like LaTeX source:

- it quotes equations as written, without re-checking them against page images;
- it cites pages from the transcription's page markers, including printed page numbers;
- it opens the figures PDF only for questions about what a figure shows;
- it escalates only when the transcription is contradicted by a derivation or a companion paper, is internally inconsistent, or is structurally defective. Claude never corrects a transcription silently.

3 What it costs

	Reading the PDF directly	**Reading a transcription**
One-time	none	conversion: about 2,500–3,000 tokens per page, plus about 1,500 per proofreading sheet (four equations each)
Each read of a page	a page image (up to about 1,600 tokens) plus crops of the relevant lines, often several thousand tokens in total	the page's Markdown text, typically well under 1,000 tokens
Each new session	the same again	the same text cost; no images

Per-read figures are approximate and depend on page density and on how many lines need cropping.

A page costs roughly as much to convert as one or two careful image readings of it. Converting therefore pays for itself whenever a document will be consulted for more than a handful of equations, or across more than one session. After that, every read is cheaper.

4 When to convert

Convert equation-heavy papers, lecture notes and textbook chapters that you will keep returning to, especially documents stored in a Project.

Leave as PDF in these cases:

- a one-off question about a single equation, where direct reading is cheaper;
- figure-centric documents, since text inside figures is not transcribed;
- documents whose layout itself is the question.

Convert whole books one chapter per conversation.

5 Workflow

1. **Convert.** In a chat, attach the PDF and ask Claude to convert it to Markdown. For long documents, give a page range.
2. **Download** `NAME.md` and, if present, `NAME_figures.pdf`.
3. **Upload** both to the Project's knowledge. The figures PDF can be omitted if no figure will need to be seen, because every caption is already in the Markdown.
4. **Keep or remove the original PDF.** If both are present, Claude reads the transcription for content and opens the PDF only for questions about the page itself.
5. **Update the protocol**, if the Project has one. The staleness check reports the document as *converted*; accept Claude's offer to rewrite its entry.

> **Where the trust lies.** A transcription moves verification from reading time to conversion time. Displayed equations are proofread against the PDF. Inline math is checked by re-reading the strips with mandatory zooms on accents and indices, not by side-by-side sheets. Any symbol left ambiguous at conversion is named in the conversion report: keep that report.